

Seokyung Park

M.S. Student · Agricultural Robotics & Smart Sensing

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Research interests

My research combines agricultural sensing, imaging and plant phenotyping. I use hyperspectral imaging to assess tomato damage, map greenhouse microclimates and develop quadruped-based 3D phenotyping. I aim to quantify crop traits and their relationships with growing conditions through integrated sensing and robotic platforms.

Education

Pusan National University

Mar. 2025 – Feb. 2027 (expected)

M.S. in Bio-Industrial Machinery Engineering | GPA: 4.36/4.50

Pusan National University

Mar. 2017 – Feb. 2025

B.S. in Plant Bioscience | GPA: 4.26/4.50

Minor in Bio-Industrial Machinery Engineering

Summa Cum Laude; President's List

Research experience

Graduate Researcher

Mar. 2025 – Present

Precision Agriculture and Smart Sensing System Lab., Pusan National University

Damage-aware robotic gripping for tomato harvesting

- Developed Python software for compression experiments and force-signal processing with a myCobot 320 Pi, adaptive gripper and force sensor; compared four loading rates (6.9–15.3 N/s).
- Combined force–deformation analysis and hyperspectral imaging at 1 hour and 5 days after compression to assess delayed damage; contributed to experimental design, analysis and writing as first author.

Quadruped robotics for agricultural sensing

Lab project participant: NRF/MSIT, Grant RS-2026-25475333.

Individual research grant recipient: NRF/MOE, Grant RS-2025-25438074 (see Research funding).

- Co-first author: mapped greenhouse temperature, relative humidity and CO₂ using a Unitree Go2 at two canopy heights across eight lanes; produced 24 maps to assess fixed-sensor representativeness.
- Current M.S. thesis: developing quadruped-based 3D plant phenotyping.

Quadruped-based upstream reservoir surveys

Contract research for the National Disaster Management Research Institute (NDMI). Status: Completed.

- Conducted quadruped-based upstream soil surveys under the project on integrated reservoir embankment risk prediction and maintenance technologies (Phase I); combined RGB imaging and RTK-GNSS into location-linked datasets.

Air-dome foundations and structural sensing

Both studies below: IPET/MAFRA lab project, Grant RS-2025-02313370 (Agri-Food Export Enhancement Technology Development Program).

- **Uplift resistance of expandable foundation piles.** Modelled wing inclination and dimension effects in Abaqus/Standard; analysed uplift load–displacement behaviour and soil plastic zones in sandy soil.
- **Load-cell sensing for air-supported domes.** Installed load cells and acquired cable-tension data at Damyang (15 June 2026), using an indicator, USB-to-RS485 interface and Raspberry Pi.

Research experience (continued)

Undergraduate Research Student

Mar. 2024 – May 2025

Functional Crop Laboratory, Pusan National University

- Worked on soybean leaf phenotyping under drought using time-series RGB images, YOLOv8, Roboflow and ImageJ, with Arduino-based environmental measurements.

Publications and manuscripts

Peer-reviewed publication

- [1] J. Park, S. Park, E. Jung, J. Song, G. Park, J. Lee, M. Kim, and J. Park. Inertial sensor applications in agriculture: Sensor types and data utilization techniques for mechanical and wearable applications. *Korean Journal of Agricultural Science*, 52(4):447–469, 2025. doi: 10.7744/kjoas.520405.

Under review

Park, S., Song, J., Hwang, H., Park, G., and Park, J. (2026). Speed-dependent control of robotic gripper for tomato harvesting to minimise compression-induced damage. *Biosystems Engineering*. **Under review.**

Manuscripts in preparation

Hwang, H.[†], **Park, S.**[†], and Park, J. Greenhouse microclimate mapping using a quadruped robot to assess fixed sensor representativeness. [†]Equal contribution.

Park, S., Park, J., Jang, J., Park, D., Kwon, S., Kim, M., and Choi, G. Analysis of uplift resistance characteristics of expandable piles in sandy soil with varying wing inclination and dimensions.
English translation of the Korean manuscript title.

Research funding (individual award)

- **Individual research grant:** National Research Foundation of Korea (NRF), funded by the Ministry of Education (MOE). Grant **RS-2025-25438074**.
Development of Locomotion Algorithms and Autonomous Control Systems for Quadruped Robots Specialized for Korean Agricultural Terrain.

Technical skills

- **Programming and data analysis:** Python, NumPy, pandas and SciPy. Sensor time-series processing, force–deformation analysis, nonlinear model fitting and research visualisation.
- **Robotics and mobile sensing:** ROS 2, Unitree Go2, RTK-GNSS and LiDAR/IMU. Timestamped robot/sensor recording, greenhouse microclimate mapping, and matching ground images to GNSS positions and robot travel routes.
- **Instrumentation and data acquisition:** Force sensors, load cells, Raspberry Pi, Arduino and RS485. Force measurements for robotic compression tests, environmental measurements and field load-cell acquisition with CSV logging.
- **Imaging and plant phenotyping:** RGB and hyperspectral imaging; SAM, YOLOv8, Roboflow and ImageJ. Fruit-region segmentation and spectral comparison; soybean leaf-image analysis under drought stress and location-linked ground-image datasets.
- **Finite-element modelling:** Abaqus/CAE and Abaqus/Standard. Three-dimensional pile–soil contact models, displacement-controlled uplift simulations, geometric parameter studies and soil plastic-zone interpretation.

Selected conference presentations

- **Park, S.** (14 July 2026). Height-resolved 3D microclimate mapping and spatiotemporal analysis within greenhouse crop canopies using a quadruped robot. *AKABFE Student Research Spotlight, ASABE Annual International Meeting*, Indianapolis, USA. Presenter.
- **Park, S.**, Park, J., Hwang, H., and Kang, S. (2026). Ground-reaction-force-based lateral slope estimation for furrow-following quadruped robots. *Korean Society for Agricultural Machinery Spring Conference*.
- **Park, S.**, Park, J., Song, J., and Kim, M. (31 October 2025). Analysis of pile pull-out behaviour in Middle Eastern soil using digital image correlation. *Korean Society of Agricultural Engineers Autumn Conference*, Jeju.
- Hwang, H., **Park, S.**, et al. (22 May 2025). Remote control and integrated sensor-data environment for an agricultural quadruped robot. *Korean Society for Precision Agriculture Spring Conference*. Co-author.
- **Park, S.**, Park, J., Park, G., Song, J., and Hwang, H. (25 April 2025). Minimising damage and optimising control in fruit and vegetable harvesting using a robotic arm and gripper with real-time force sensing. *Korean Society for Agricultural Machinery Spring Conference*.

English translations are used for Korean presentation titles.

Honours and awards

- **Best Paper Presentation Award (team)**, Korean Society for Precision Agriculture, May 2025 and November 2024.
- **Excellent Paper Presentation Award**, Korean Society for Agricultural Machinery, April 2025 and October 2024.
- **Excellence Award**, PNU–PKNU Field Problem Research Group Final Symposium, November 2024.
- **Best Presentation Award (team)**, Korean Society for Engineering Education, September 2024.
- **Bronze Prize**, PNU College of Natural Resources and Life Science Capstone Design Competition, October 2023.
- **Excellence Award**, Busan LINC 3.0 Entrepreneurship Shared University Competition, January 2023.
- **PNU merit scholarships**: four full-tuition and two partial-tuition awards, 2017–2024.

Selected coursework

Graduate: Biological Image Processing; Statistical Processing and Data Analysis in Bio-Industrial Machinery Engineering; Advanced Bio-Industrial Machinery Design; Biosystems Mechanics; Advanced Biomaterials Engineering I.

Undergraduate – engineering and sensing: Biosensor and Bio-Instrumentation & Practice; Biosystems Control Engineering; Bio-Production Mechatronics Design; Applied Mechanics of Materials; Bio-Industrial Machine Element Design (I); Food Process Engineering & Practice.

Undergraduate – programming and quantitative methods: Basic Computer Programming; Experimental Design for Biologists; Biostatistics; Calculus (II).

Undergraduate – plant science and cultivation: Plant Physiology and Practice; Plant Genetics; Plant Molecular Biology; Plant Biotechnology; Plant Breeding (I, II); Soil Science; Plant Ecology; Protected Horticulture.

Additional training

Modeling Contact with Abaqus/Standard, VPK PLM Solution Training Center, 25–26 June 2026.

Introduction to Abaqus/CAE, VPK PLM Solution Training Center, 18–20 March 2026.